

Modern vehicles rely on a distributed electronic architecture composed of multiple interconnected control systems.

## Vehicular Communication Networks

### Electronic Control Unit

It functions as a small embedded computer, acquiring information from sensors, processing this data in real time, and controlling actuators.

The development of ECUs became increasingly relevant during the 1970s and 1980s, driven by the need to improve engine performance and emissions control.

Before the development of ECU, automotive systems were mostly mechanical. As vehicles became more complex, the need for a centralized control system grew.

### Controller Area Network

The CAN Network is a vehicle message-based communication protocol designed to allow an ECU to communicate to other ECUs in the vehicle.

This protocol was developed by Robert Bosch GmbH in the 1980s as a response to the increasing complexity of vehicle electronics.

[\[H\] \[width=0.8\]images/Can\\_protocol.png](#) [CAN Application Example](#) [?]

CAN protocol is a message-based protocol. This means the data of a vehicle is transmitted in messages identified by a unique identifier.

These messages, known as frames, follow a specific format as to ensure consistency:

- ID - Unique message identifier
- DLC (Data Length Code) - Specifies the data length in bytes
- Data Field - Contains the information payload
- CRC (Cyclic Redundancy Check) - Used for error and data integrity verification
- ACK Slot - Confirms reception of at least one node
- EOF (End of Frame) - Signals the end of the message

[\[H\] \[width=1\]images/CAN-Protocol-Extended-Frame-Format.png](#) CAN Standard Data Frame [?]

### Controller Area Network Data Base

CAN databases, commonly represented as DBC files, play a fundamental role in the interpretation and use of CAN messages.

definitions without requiring any modification to the stored dataset. CAN databases are also widely used in simulation and testing.

### Why CAN is the Preferred Method for Intra-Vehicular Communication

Controller Area Network (CAN) is widely adopted in intra-vehicular communication due to its deterministic behavior and reliability.

CAN employs a multi-master, message-oriented communication model over a differential two-wire bus, which increases robustness against electromagnetic interference.

A key feature of CAN is its non-destructive bitwise arbitration mechanism, which guarantees deterministic access to the bus.

In terms of efficiency, CAN minimizes wiring complexity by using a shared bus topology, reducing both system cost and weight.

### Standalone ECU

Standalone ECUs are aftermarket electronic control systems designed to fully replace the original equipment manufacturer (OEM) ECUs.

Unlike OEM ECUs, which are typically optimized for emissions, reliability, and mass production constraints, standalone ECUs are designed for performance and customization.

The motivation for using standalone ECUs arises from the need for adaptability and transparency. In motorsport and racing, drivers often want to monitor and adjust engine parameters in real-time.

### Speeduino/RusEFI ECUs

Speeduino and RusEFI are open-source standalone ECU platforms designed to provide flexible and low-cost engine management solutions.

Speeduino is based on Arduino-compatible hardware and focuses on simplicity and accessibility. It provides core engine control functions like fuel injection and ignition timing.

RusEFI, on the other hand, targets more advanced use cases with a higher level of performance and configurability. It supports more complex engine configurations and tuning options.

The main motivation behind both platforms is to provide open, customizable alternatives to proprietary ECUs. This allows enthusiasts and racers to tailor their engine management systems to their specific needs.

### Embedded System's Development Platforms

#### Raspberry Pi Ecosystem

For the controller of our project, we decide to utilize a Raspberry Pi Zero 2W [?]. This decision was made due to several factors:

- Full Linux Computer:** The controller will have to implement several different tasks at once such as:
  - Real-time data acquisition from the ECU
  - Data processing like filtering and parsing signals
  - Database logging and management
  - UI rendering
  - Real-time bluetooth data transfer
- A Raspberry Pi system provides a good flexibility to develop our application while being able to build our application with various languages.
- Fast processor and decent memory:** The Raspberry Pi Zero 2 W incorporates a quad-core 64-bit Arm Cortex-A53 processor, providing sufficient processing power for the tasks at hand.
- Easy communication with hardware:** The Raspberry Pi provides a USB serial port with the possibility to use UART for direct communication with the ECU.
- Networking:** This computer provides a Bluetooth Low Energy (BLE) system which allows the transfer of the CAN signal data to a mobile application.

With all this said, we choose the Raspberry Pi for its fast performance, and affordable price. Another deciding factor was the availability of development resources.

#### Utilized Programming Languages

Support for multiple programming languages was a key design consideration, allowing each component to leverage the strengths of its preferred language.

- Python:** Used for the controller application for a ECU data pipeline, controller logic and bluetooth integration due to its ease of use and extensive libraries.
- SQLite:** Although Python is used for making requests to the database, said requests are in SQL using a SQLite variant for local storage.
- Kotlin:** Kotlin was chosen as the primary language for mobile application development due to its seamless integration with Android and its concise syntax.

### Data Analysis Techniques in Motorsport

Data analysis plays an important role in motorsport, where vehicle performance and reliability depend on the ability to interpret large amounts of data.

The analysis process generally begins with data preparation. Measurements acquired from different sensors may come in various formats and need to be cleaned and organized.

After preparation, vehicle data can be examined using statistical and time-domain analysis. Statistical methods provide a high-level overview of the data, while time-domain analysis allows for more detailed inspection of specific events.

#### Statistical Analysis

Descriptive statistics provide an initial overview of the collected data. Measurements such as the minimum, maximum, and mean are essential for understanding the range and central tendency of the data.

The mean represents the average operating value of a signal, while the minimum and maximum identify its observed extremes.

Statistical analysis may be performed over an entire acquisition session or over selected time intervals. Whole-session analysis provides a broad overview, while interval-based analysis allows for more granular investigation.

In motorsport applications, the interpretation of a single parameter is often insufficient. Several signals must be compared to gain a comprehensive understanding of vehicle performance.

#### Session Comparison

Comparing multiple acquisition sessions is useful for evaluating changes in vehicle configuration, environmental conditions, and driver performance.

For a meaningful comparison, equivalent portions of each session should be selected whenever possible. Differences in track conditions or vehicle wear can affect the data, so comparing similar laps or segments is crucial.

Session comparison supports iterative vehicle development by providing objective evidence about the effect of technical changes.

#### Anomaly Identification

Anomaly identification consists of detecting measurements or patterns that differ from the expected behaviour of the system.

Statistical thresholds may also be used when fixed limits are not sufficient. Measurements that deviate significantly from the mean or exceed predefined limits can indicate potential issues.

For this reason, anomaly detection should consider the duration and context of the event. A condition that persists over time is more likely to be a genuine problem than a transient spike.

The purpose of anomaly identification in the proposed platform is not to perform safety-critical diagnosis or automatic control, but to alert the user to potential issues that require further investigation.

#### Visualization of Acquired Data

Visualization complements statistical analysis by showing how vehicle parameters evolve over time. Time-series plots and graphs are essential for understanding the dynamics of the data.

Dashboards provide an immediate representation of current operating conditions, while session plots and statistical summaries allow for deeper analysis.

Effective visualization should present the most relevant information without overwhelming the user. Signals should be clearly labeled, and the interface should be intuitive and easy to navigate.